Ben Parrack

(804) 221-0732 | benparrack@gmail.com | LinkedIn | GitHub

EDUCATION

Virginia Tech | Blacksburg, VA **Expected May 2028**

Bachelor of Science in Computer Science ***GPA: 3.5/4.0***

Major in Data-Centric Computing, Minors in Mathematics, Artificial Intelligence, and Biomedical Engineering

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Computer Organization, Data Analytics and Visualization, Applied Combinatorics, and Linear Algebra
- **Honors:** Dean's List (every semester)

TECHNICAL SKILLS

Languages: Java, Python, C, Bash, Ladder Logic

Tools and Technologies: Git, Linux, PyTorch, VMware, RSLogix, Arduino, OpenCV, MediaPipe, SOLIDWORKS, MATLAB

Certifications and Training: AWS GenAI Training and Jam: Amazon Bedrock and SageMaker (Sep 2025)

EXPERIENCE

GrayMatter Systems | Ashland, VA | *Automation Intern* **May - Aug 2026**

- Installed PLCs at 50+ sites, rewired them inside the panels, and updated the Ladder Logic programs
- Configured PLC network settings via Rockwell software, allowing communication with SCADA systems
- Began billable client work within 2 weeks of starting site work
- Designed and implemented an inventory management system for GrayMatter's warehouse according to specified constraints (flexible, easily updatable, M365-compatible)

PROJECTS

Live ASL Recognition | *Personal Project* **Aug 2026 - Present**

- Built a program that translates American Sign Language via webcam in real time with MediaPipe and OpenCV
- Trained 4 different models on an ASL alphabet dataset, and a hand-landmark MLP was the most accurate
- Added multiple word datasets and reached 93% accuracy on ~250 words

LLM-Assisted Evaluation of Classroom Layouts and Movement Behaviors | *Research* **Jan - May 2026**

- Reviewed minute-long segments of structured learning environment videos and compared LLM findings to the actual scenarios, identifying key differences to formulate improvements for the model's behavior
- Leveraged analysis to refine prompts for classifying different behaviors and postures, with the end goal of improving student attentiveness

Interdisciplinary Project (IDPro@VT) - Hydroponics Design Team | *Electronics Lead* **Aug - Dec 2025**

- Developed a hydroponics control system that integrated Arduino-based sensor monitoring and pump control to regulate nutrient delivery for optimized growth
- Configured instrumentation such as pH sensors and TDS probes to output quantitative values in Arduino IDE
- Created a project summary poster and presented with teammates to 50+ people

VT Gym Occupancy Graph Program | *Personal Project* **Sep 2025**

- Built a Python web scraper that pulled gym occupancy data every 5 minutes and visualized trends with matplotlib to help students optimize the timing of gym visits

VOLUNTEER EXPERIENCE

CS^2 (Computer Science Community Service) | Blacksburg, VA | *Executive Board Member* **Aug 2025 - Present**

- Led a program that taught 8 middle schoolers how to use Arduinos by guiding them through the process of building a programmable remote-controlled car over the course of a 6-week project (meeting weekly)
- Designed curriculum and taught middle schoolers how to design objects in CAD and 3D print their designs